NOT MEASUREMENT SENSITIVE

MIL-G-85117A(AS)
22 AUGUST 1989
SUPERSEDING
MIL-G-85117(AS)
6 JANUARY 1983

# MILITARY SPECIFICATION

# GYRO, STORED-ENERGY, FREE

# ROLL POSITION INDICATING

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification establishes the performance, design, test, manufacture and acceptance requirements for a roll position indicating free stored-energy gyro, herein referred to as the "gyro."

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-320 Chromium Plating (Electrodeposited).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 53, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC No. N/A FSC 1420

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| | |
|---|---|
| QQ-N-290 | Nickel Plating (Electrodeposited). |
| QQ-P-35 | Passivation Treatments for Corrosion-Resisting Steel. |
| QQ-P-416 | Plating, Cadmium (Electrodeposited). |
| QQ-S-365 | Silver Plating, Electrodeposited; General Requirements for. |

MILITARY

| | |
|---|---|
| MIL-C-5541 | Chemical Conversion Coatings on Aluminum and Aluminum Alloys. |
| MIL-H-6088 | Heat Treatment of Aluminum Alloys. |
| MIL-H-6875 | Heat Treatment of Steel, Process for. |
| MIL-H-7199 | Heat Treatment of Wrought Copper-Beryllium Alloys, Process for (Copper Alloys: Numbers 170, 172 and 175). |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys. |
| MIL-T-10727 | Tin Plating; Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals. |
| MIL-C-14550 | Copper Plating (Electrodeposited). |
| MIL-C-26074 | Coatings, Electroless Nickel, Requirements for. |
| MIL-G-45204 | Gold Plating, Electrodeposited. |
| MIL-R-46085 | Rhodium Plating, Electrodeposited. |
| MIL-H-81200 | Heat Treatment of Titanium and Titanium Alloys. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-209 | Clean Room and Work Station Requirements, Controlled Environment. |



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| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-130 | Identification Marking of U.S. Military Property. |
| MIL-STD-202 | Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment. |
| MIL-STD-1523 | Age Controls of Age-Sensitive Elastomeric Materiel (for Aerospace Applications). |
| MIL-STD-45662 | Calibration Systems Requirements. |

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

| | |
|---|---|
| 1563560 | Gyroscope, Displacement, Stored-Energy. |
| 1563567 | Gyroscope, Displacement, Stored-Energy. |

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

| | |
|---|---|
| WS-6536 | Procedures and Requirements for Preparation and Soldering of Electrical Connections. |

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)



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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the document which is DOD adopted shall be that listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issue of the document not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

UNIFORM FREIGHT CLASSIFICATIONS

| | |
|---|---|
| Uniform Freight Classification 12 | Ratings, Rules and Regulations. |

(Application for copies should be made to the Uniform Classification Committee, 202 Union Station, Chicago, IL 60606.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Item description. The gyro is a roll position indicating free stored-energy gyro with two degrees of freedom. A two-wiper potentiometer provides approximate sine and cosine functions of outer gimbal rotation from the caged position. The gyro is provided with a caged and wound indicator and an inner gimbal limit indicator.

3.2 Characteristics.

3.2.1 Performance. The gyro shall perform as specified herein under the environmental conditions of 3.2.5.

3.2.1.1 Motor winding. A means shall be provided to wind the spring motor that will prevent over-wind damage to the spring while assuring full energy storage in the spring. A means shall be provided to restrain the rotor in this state which can be released by an actuation mechanism with the characteristics specified in 3.2.1.3. There shall be a means of verifying that the spring motor is fully wound prior to installation of the cover.

3.2.1.2 Caging. The gyro shall be caged in all axes and shall meet the requirements specified in 3.2.1.2.1 and 3.2.1.2.2.

3.2.1.2.1 Caged restriction. With the gyro caged and wound as specified in 3.2.1.1, the total angular play about the outer gimbal axis shall be not greater than 1.0 degree (deg) when the outer gimbal is rotated alternately clockwise (CW) and counterclockwise (CCW).



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3.2.1.2.2 Caged indication. The gyro shall be delivered caged and with the motor fully wound. Under these conditions, a resistance not greater than 1.0 ohm, measured at not greater than 10 milliamperes direct current (mAdc), shall appear between terminals 3 and 4. A resistance of not less than 10 megohms, measured at 100 volts direct current (Vdc), shall appear between terminals 2 and 4 when the gyro is caged.

3.2.1.3 Actuation. The actuation mechanism of the gyro shall have the performance characteristics specified in 3.2.1.3.1 through 3.2.1.3.4.

3.2.1.3.1 Actuator resistance. The direct current (dc) resistance between terminals 4 and 5 when measured using not greater than 10 mAdc current and the gyro at 25 ±5 degrees Celsius (°C) shall be as follows:

a. Pre-fired condition: 6.0 +1.3, -1.0 ohms.

b. Post-fired condition: Not less than 50 kilohms.

3.2.1.3.2 Actuator no-fire current. Application of 30 +0, -3 mAdc for not less than 5 minutes at +57 ±5°C between terminals 4 and 5 shall not fire the actuator.

3.2.1.3.3 Actuator firing current. A current of 300 +10, -0 mAdc applied between terminals 4 and 5 for not greater than 10 milliseconds (ms) shall cause the actuator to release the spring motor.

3.2.1.3.4 Uncaged indication. Within not greater than 100 ms from onset of the current of 3.2.1.3.3, a resistance not greater than 1.0 ohm, measured at not greater than 10 mAdc, shall appear between terminals 2 and 4. No caged indication shall recur with the gyro uncaged and operating. When the gyro is uncaged, a resistance not less than 10 megohms, measured at 100 Vdc, shall appear between terminals 3 and 4.

3.2.1.4 Runup time. The gyro shall exhibit a functional output within 100 ms from the onset of the current of 3.2.1.3.3.

3.2.1.5 Running time. When the gyro is initiated with the outer gimbal axis aligned within ±10 degrees of horizontal, the running time shall be as follows:

| Temperature Range | Running Time (minimum) seconds | Roll Rate (degrees per second) | Direction |
|---|---|---|---|
| 15 to 57°C | 125 | 75 | CW/CCW |
| 15 to 75°C | 100 | 260 | CW/CCW |
| 15 to -43°C | * | 75 | CW/CCW |
| 15 to -43°C | ** | 260 | CW/CCW |

* Running time requirements shall decrease linearly in proportion to temperature from 125 seconds to not less than 100 seconds.

**Running time requirements shall decrease linearly in proportion to temperature from 100 seconds to not less than 75 seconds.



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3.2.1.6 Gimbal freedom.

3.2.1.6.1 Inner gimbal freedom. The inner gimbal shall have 85 ±2 degrees of angular freedom in either direction about the inner gimbal axis.

3.2.1.6.1.1 Inner gimbal lock indication. When the inner gimbal is within the limits of angular motion of 3.2.1.6.1, a resistance not less than 10 megohms, measured at 100 Vdc, shall appear between terminals 1 and 4. The operating acceleration specified herein shall not cause the resistance to drop below 10 megohms at any time. When the inner gimbal reaches either limit of angular motion of 3.2.1.6.1, a momentary resistance not greater than 1.0 ohm, measured at not greater than 10 mAdc, shall appear between terminals 1 and 4; this indicates gimbal lock.

3.2.1.6.2 Outer gimbal freedom. The outer gimbal shall have continuous 360 degrees of angular motion in either direction about the outer gimbal axis.

3.2.1.7 Pickoff potentiometer. The pickoff potentiometer shall have: 1) a continuous 360 degree resistance element excited across a diameter, and the resistance element shall be fixed to the gyro case with axis colinear with the outer gimbal axis and 2) "sine" and "cosine" wipers that are orthogonal within ±4.0 degrees, and so oriented on the outer gimbal axis that the wipers meet the zero position requirements of 3.2.1.7.2 when the gyro is in the caged position. The resistance shall be compensated so that the sine and cosine approximation requirements of 3.2.1.7.2 are achieved.

3.2.1.7.1 Excitation resistance. The resistance between terminals 6 and 7 shall be 2.0 kilohms ±20 percent with terminals 8 and 9 unconnected and the gyro at 25 ±5°C.

3.2.1.7.2 Sine and cosine approximation. With the pickoff potentiometer resistive element excited, as shown on Figure 1, the voltage outputs of the sine and cosine wipers shall conform to the ideal sine or cosine waveform within ±11.0 percent of the zero to peak excitation voltage ($V_p$) at any outer gimbal angle. The zero position of the outer gimbal axis shall be defined with the output at pin 8 is 0.0 Vdc ±3.5 percent $V_p$ and the output of pin 9 is $-V_p$ ±3.5 percent $V_p$ with the gyro in the caged and wound position (see Figure 2). The wiper position when the gyro is caged and wound shall be such that when the gyro is uncaged and rotated CW (Figure 2) about the outer gimbal axis an angle $\phi$, the wiper outputs shall be $V_p \sin \phi$ ±0.110 $V_p$ on terminal 8 and $-V_p \cos \phi$ ±0.110 $V_p$ on terminal 9. Resolution and noise, from either wiper or slipring contacts, shall fall within the tolerance band, with the exception that six noise spikes will be allowed during any 360 degrees of rotation. An individual noise spike shall be not greater than 40.0 ms in duration while testing for compliance to this paragraph or to 3.2.1.8.



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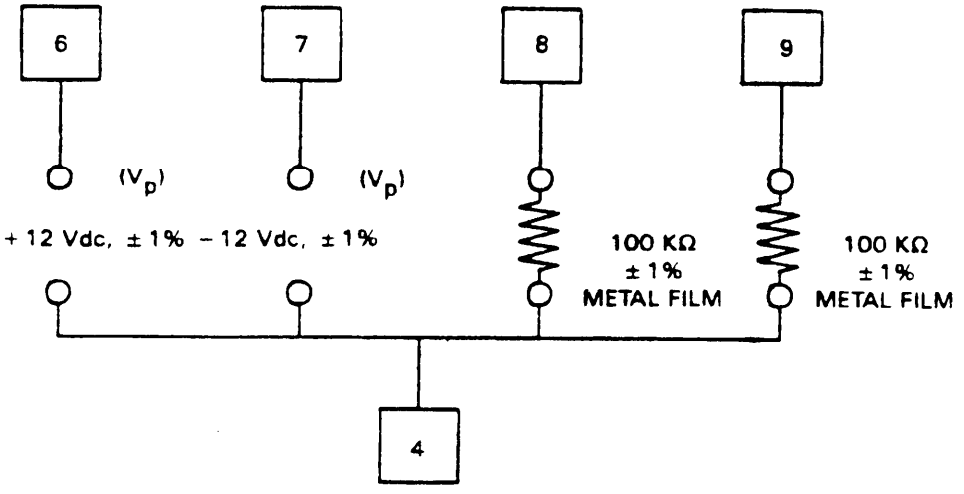



FIGURE 1. Potentiometer excitation and test circuit.

3.2.1.7.3 Uncaging offset. With the excitation and wiper loading as shown on Figure 1, the sine wiper voltage shall not change by greater than ±3.5 percent of the zero to peak excitation voltage (±0.035 $V_p$) by actuating the stationary gyro.

3.2.1.8 Drift. When tested as specified herein, the outer gimbal drift shall be not greater than 20 deg for the running time intervals of 3.2.1.5.

3.2.1.9 Electrical characteristics.

3.2.1.9.1 Dielectric withstanding voltage. With terminals 4 and 5 jumpered and only the terminal under test connected, the gyro shall without disruptive discharge or impairment of subsequent function withstand 200 volts root-mean-square (Vrms), 60 Hertz (Hz), applied for not less than 60 seconds in first article and lot testing, and 1 to 3 seconds in quality conformance testing, between each gyro terminal and the gyro case when tested as specified herein.



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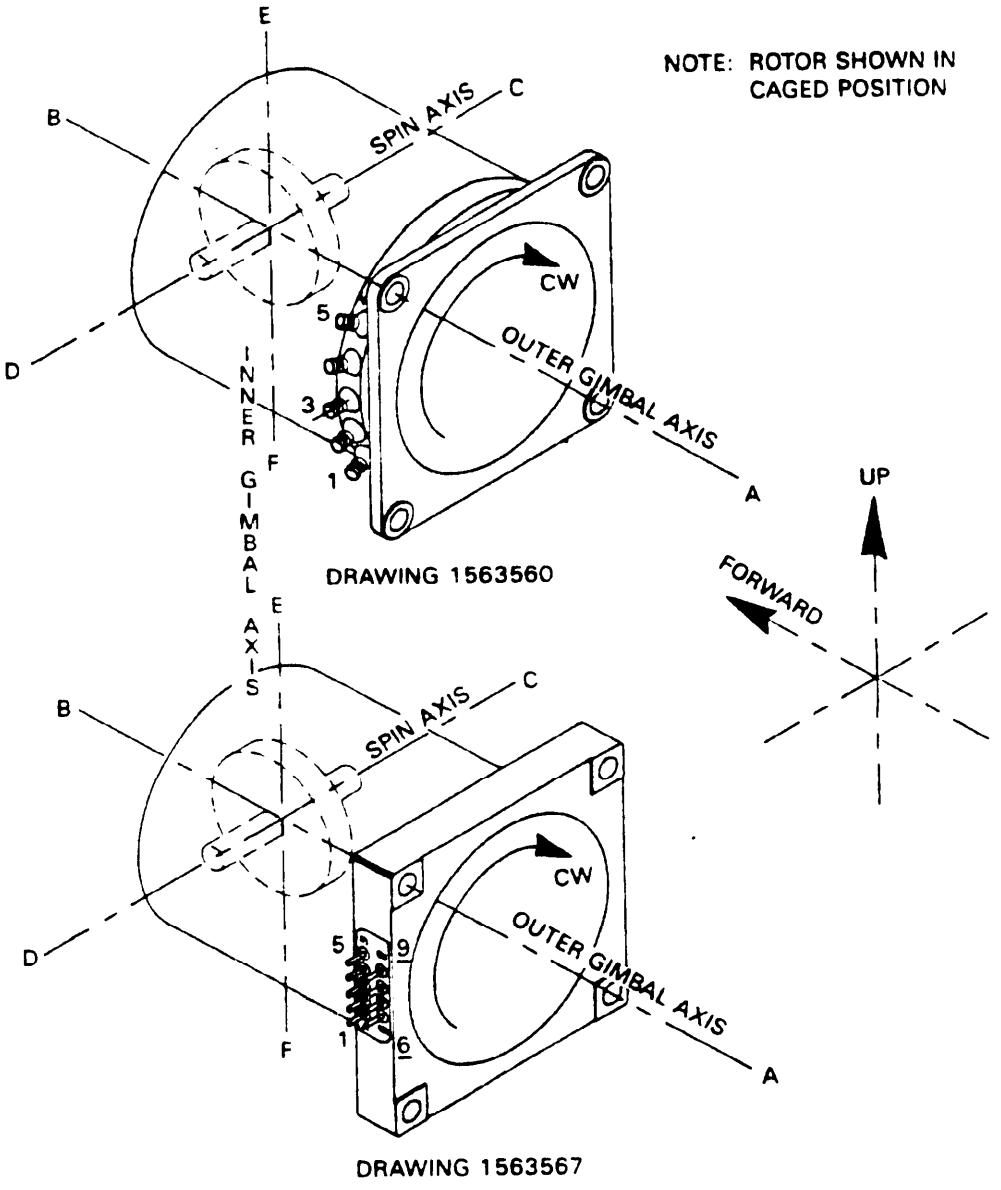



FIGURE 2. Gyro reference axes.

3.2.1.9.2 Insulation resistance. With terminals 4 and 5 jumpered, insulation resistance between electrically isolated interior circuits and between each such circuit and the gyro case shall be not less than 10 megohms when measured as specified herein.

3.2.1.9.3 Actuator insulation resistance. After firing as specified in 3.2.1.3.3, the resistance between terminal 5 and the gyro case shall be not less than 50 kilohms.



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3.2.1.10 Terminal or header pin strength. Terminals (Drawing 1563560) or header pins (Drawing 1563567) shall meet the pull strength test requirements of the applicable part drawing.

3.2.2 Physical characteristics.

3.2.2.1 Weight. The weight of the gyro shall not be greater than 18 ounces.

3.2.2.2 Dimensions. Gyro outline and mounting dimensions shall be not greater than the limits specified on the applicable part drawing.

3.2.2.3 Filling and leakage. The interior of the gyro as delivered shall be filled with a mixture of 10 ±1 percent helium and 90 ±1 percent dry nitrogen by volume. The mixture shall be free of particles larger than 0.1 micron in maximum dimension, and shall contain not greater than 0.006 milligram of water vapor per liter (dew-point -65°C) at the filling pressure. The filling pressure shall be 14.7 ±0.5 pounds per square inch absolute (psia). Leakage detected during fine leak test shall be not greater than $5 \times 10^{-7}$ atmospheric cubic centimeter per second at not less than 1.0 atmosphere pressure differential when tested as specified herein.

3.2.3 Reliability. The design reliability of the gyro shall be established by the selection and inspection of materials, parts, and processes used in fabrication and shall be 0.999 which shall be demonstrated by prediction. The performance of the gyro shall be sufficiently reliable to successfully meet all requirements of first article inspection. Production reliability demonstration shall be 0.843 at 80 percent confidence.

3.2.3.1 Life. Gyro design shall be capable of not less than 12 actuations with rewind and squib replacement in accordance with 3.2 for gyro operational characteristics.

3.2.4 Maintainability. The gyro shall be designed so as to require no periodic maintenance.

3.2.5 Environmental conditions. The gyro shall meet the requirements specified in 3.2.1 after exposure to the nonoperating environments, and during and after exposure to the operating environments specified herein.

3.2.5.1 Nonoperating environments.

3.2.5.1.1 Temperature cycle. The gyro shall perform as specified in 3.2.1 after being subjected to temperature cycling at +57 ±5°C and -43 ±5°C.

3.2.5.1.2 Sine vibration. The gyro shall perform as specified in 3.2.1 after being subjected to a sine vibration frequency at a logarithmic rate from 5 to 500 to 5 Hz in 15 minutes in the manner specified with the gyro at a temperature of -43 ±5°C and +57 ±5°C.



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3.2.5.1.3 Random vibration. The gyro shall perform as specified in 3.2.1 after being subjected to random vibration of 20 to 2,000 Hz in the manner specified with the gyro at a temperature of -43 ±5°C and +57 ±5°C.

3.2.5.1.4 Temperature shock and altitude. The gyro shall perform as specified in 3.2.1 after being subjected to six temperature shock cycles ranging from +71 ±5°C at ambient barometric pressure to -43 ±5°C at a 50,000-foot barometric altitude.

3.2.5.1.5 Handling shock. The gyro shall perform as specified in 3.2.1 after being subjected to a total of 36 handling shock cycles in the manner specified herein with the gyro at a temperature of -43 ±5°C and +57 ±5°C.

3.2.5.1.6 Humidity. The gyro shall perform as specified in 3.2.1 after being subjected to a relative humidity of 95 percent.

3.2.5.2 Operating environments.

3.2.5.2.1 Temperatures. The gyro shall perform as specified in 3.2.1 when actuated at any gyro temperature from -43 ±5°C to +57 ±5°C.

3.2.5.2.2 Acceleration. The gyro shall perform as specified in 3.2.1 when exposed to 25 ±3 gravity units (g) acceleration applied along the A-B axis only (Figure 2), in the direction A to B, for not less than 5 seconds starting at the onset of the voltage of 3.2.1.3.3.

3.2.5.3 Storage life. The gyro shall be designed to withstand 5 years of sheltered storage and then meet the requirements specified herein.

3.2.6 Transportability. Transportation requirements shall be in accordance with the Uniform Freight Classification 12 which permits shipment under the minimum requirements for a device of this type.

3.3 Design and construction. Gyro design and construction shall conform to Drawing 1563560 or 1563567 and comply with 3.3.1.

3.3.1 Materials, processes and parts. Materials, processes and parts to which particular MIL-STD-454 requirements pertain shall, wherever possible, comply with those requirements.

3.3.1.1 Materials.

3.3.1.1.1 Fungus resistance. Only materials of MIL-STD-454, Requirement 4, Group I (fungus-inert in all states and grades), shall be exposed on the gyro exterior. Other materials are permissible in the hermetically sealed interior.

3.3.1.1.2 Age-sensitive elastomers. The gyro shall incorporate no elastomers of the type to which MIL-STD-1523 applies age controls.



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3.3.1.1.3 Lubricants. Gyro lubricants shall be consistent with the storage life requirements of 3.2.5.3.

3.3.1.1.4 Adhesives. Any adhesive within or on the exterior of the gyro shall comply with MIL-STD-454, Requirement 23.

3.3.1.1.5 Dissimilar metals. Dissimilar metals shall not be used in direct contact unless protected against electrolytic corrosion. Selection and protection of dissimilar metal combinations shall meet the provisions of MIL-STD-454, Requirement 16.

3.3.1.1.6 Magnesium. The gyro shall incorporate no pure magnesium nor alloy in which magnesium is the major constituent.

3.3.1.2 Processes.

3.3.1.2.1 Brazing. Any brazing in gyro fabrication shall comply with MIL-STD-454, Requirement 59.

3.3.1.2.2 Welding. Any structural welding in gyro fabrication shall comply with MIL-STD-454, Requirement 13.

3.3.1.2.3 Heat treatment. Any heat treatment in gyro fabrication shall comply with MIL-H-6088 for aluminum alloys; MIL-H-6875 for steels; MIL-H-7199 for beryllium copper; or MIL-H-81200 for titanium.

3.3.1.2.4 Plating. Any plating in gyro fabrication shall be by processes minimizing long-term flaking and particulate contamination. Plating shall comply with QQ-C-320, Class 2, Type II, for chromium; QQ-N-290, Class 1, or MIL-C-26074, Class 1 or 2, for nickel; QQ-P-416, Class 2, Type II, for cadmium, with no cadmium to be used in contact with titanium or on parts on moving contact; QQ-S-365, Type II, Grade B, for silver; MIL-G-45204, Type II, for gold; MIL-R-46085, Type I, Class 2, for rhodium; MIL-C-14550 for copper; or MIL-T-10727 for tin. Zinc plating shall not be used.

3.3.1.2.5 Passivation. Corrosion and heat resistant steels in the gyro shall be passivated in accordance with QQ-P-35.

3.3.1.2.6 Aluminum surfaces. Aluminum or aluminum-alloy parts in the gyro shall be anodized in accordance with MIL-A-8625 or chemically coated in accordance with MIL-C-5541, with subsequent painting permissible.

3.3.1.2.7 Soldering. Soldering of electrical conductors in the gyro shall comply with WS-6536.

3.3.1.3 Parts.

3.3.1.3.1 Springs. Springs in the gyro shall comply with MIL-STD-454, Requirement 41.



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3.3.1.3.2 Bearings. Bearings in the gyro shall comply with MIL-STD-454, Requirement 6.

3.3.1.3.3 Castings. Castings in the gyro shall comply with MIL-STD-454, Requirement 21.

3.3.2 Identification and marking. The gyro shall be identified in accordance with MIL-STD-130 by a marking that will not deteriorate under environmental conditions of 3.2.5. At minimum, the marking shall include the proper Part Number, "1563560" or "1563567", and revision letter; a unique serial number; manufacturer's name or symbol and Federal Supplier Code, if any; manufacturer's part number; and date of manufacture. Terminals or header pins shall be labeled "1", "5", "6" and "9".

3.3.3 Interchangeability. Gyro interchangeability shall be in accordance with MIL-STD-454, Requirement 7.

3.3.4 Standards of manufacture.

3.3.4.1 Cleanliness. All critical gyro assembly operations shall take place in a clean environment approved by the contracting activity or certified to FED-STD-209, Work Area, Class 1000 requirements. All components and subassemblies shall be inspected for cleanliness before final assembly.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.2.1c and 6.2.3).

3.5 Workmanship. Workmanship shall comply with MIL-STD-454, Requirement 9.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or



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quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Lot inspection (see 4.5.2)

4.3 Inspection conditions.

4.3.1 Inspection equipment, facilities, and conditions. Inspection equipment and facilities shall be adequate in quality and quantity to permit performance of all examinations and tests specified herein. Environmental conditions not otherwise specified herein shall be within the following limits during inspections.

| | | |
|---|---|---|
| a. | Air temperature: | 15 to 35°C. |
| b. | Relative humidity: | 10 to 95 percent. |
| c. | Barometric pressure: | Ambient, at ground elevation shall be not greater than 7,000 feet. |

4.3.2 Clean room. Removal of the gyro cover, gyro operation with cover removed (as for measurement of runup time), actuator replacement, motor rewinding, and cover replacement, shall take place only in a clean environment meeting the requirements of 3.3.4.1.

4.3.3 Test equipment accuracy and calibration. Test equipment aggregate error (from inaccuracy, limitation of resolution, and normal error of perception) shall be not greater than one-tenth of the tolerance upon the parameter tested; or in the instance of a single-limit tolerance, the error shall be applied to the observation in such direction as to assure that the parameter satisfies the limit. Test equipment calibration shall be in accordance with MIL-STD-45662.



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4.3.4 Tolerances on specific test conditions. Unless otherwise specified herein, the maximum allowable tolerances on test conditions specified herein, exclusive of equipment accuracy, shall be as follows:

| | | |
|---|---|---|
| a. | Temperature: | ±2°C. |
| b. | Altitude | ±5 percent of pressure value. |
| c. | Relative humidity: | +5, -0 percent. |
| d. | Vibration acceleration: | |
| | (1) Sinusoidal: | ±10 percent of peak. |
| | (2) Random | Not greater than 1.5 decibel (dB) of power spectral density or +40, -30 percent root-mean-square (rms). |
| e. | Vibration frequency: | ±2 percent. |
| f. | Static loads: | +2, -0 percent. |
| g. | Shock loads: | +0, -10 percent. |
| h. | Shock duration: | ±10 percent. |
| i. | Voltage, alternating current (ac), rms, or dc: | ±1 percent. |
| j. | Voltage, ac (peak-to peak): | ±5 percent. |
| k. | Current, dc: | ±1 percent. |
| l. | Frequency: | ±1 percent. |
| m. | Time: | ±1 percent. |
| n. | Resistance (dc): | ±1 percent. |

4.3.5 Temperature stabilization. When tests are required at low or high temperatures, the gyro shall be conditioned at the temperature specified herein for not less than 1 hour prior to testing. If the gyro is removed from the temperature chamber for performance of the tests, all testing shall be completed within 20 minutes after removal. If the tests are not completed within the 20 minute period, the gyro shall be returned to the temperature chamber for 1 hour before continuing the tests.



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4.4 First article inspection. When specified in the contract or purchase order (see 6.2.1), a first article inspection shall be performed in accordance with Table I. Unless otherwise specified in the contract or purchase order, the first article sample shall consist of 16 gyros which shall be manufactured using the same design, materials, parts, and processes proposed for production. Eight gyros shall be randomly selected from the first article sample and subjected to the inspections specified in Table I. Unless otherwise specified in the contract or purchase order, any gyro design change may require the first article inspection to be repeated. Any production prior to approval of the first article sample shall be at the contractor's risk.

4.4.1 Action in event of first article inspection failure. Failure of one or more gyros of the first article sample to meet all requirements of the Table I shall be cause for rejection of the sample. The contractor shall then take action acceptable to the contracting activity to correct the cause(s) of failure. No first article sample lot of gyros will be considered for acceptance until the lots reflects corrective action or until a new first article sample reflecting this action passes first article inspection.

4.4.2 Disposition of first article samples. Disposition of first article test samples procured under this specification shall be as specified in the contract or purchase order (see 6.2.1).

4.5 Quality conformance inspection (QCI). QCI shall verify that all requirements of Sections 3 and 5 have been met. Every gyro shall be subjected to the quality conformance inspection of Table II, with procedural sequence as shown for Group 1 characteristics and sequence optional for Groups 2 and 3.

4.5.1 Action in event of quality conformance inspection failure. Failure of a gyro to meet all Table II, Group 1, requirements, or all Group 2 requirements for which the gyro is individually inspected, shall because for rejection of that gyro only. A gyro rejected for failure to meet individual requirements may be reworked and resubmitted for acceptance. Failure of a production lot of gyros to meet all Group 3 requirements for which the lot is inspected collectively, by examination of manufacturing data or certifications, shall be cause for rejection of the lot. With contracting activity approval, a lot so rejected may be reworked and resubmitted for acceptance. Failure of packaging/packing of a production lot of gyros to meet all Group 5 requirements shall be cause for rejection of the lot. With contracting activity approval, a lot so rejected may be repackaged/repacked and resubmitted for acceptance, subject only to Group 5 reinspection and reinspection for the Group 2 requirement of presence of caged and uncaged indication and Group 3 requirements for Identification, Workmanship and Cleanliness.



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TABLE I. First article inspection.

| Inspection requirement 5/ | Conditions and reference paragraph (4.6) | Requirement paragraph | Inspection paragraph | Specimen Nos. 1/ 3/ |
|---|---|---|---|---|
| Tables II, Groups 1, 2, 3, and 4 | Ambient conditions 2/ (4.3) | Table II | Table II | 1 - 7 |
| Temperature cycling and vibration: | (Nonoperating preconditioning) 4/ | | | |
| Temperature cycle | | 3.2.5.1.1 | 4.6.10.1.1 | 1 - 7 |
| Sine vibration | +57°C, Axes A-B, C-D, E-F | 3.2.5.1.2 | 4.6.10.1.2 | 1 - 7 |
| Sine vibration | -43°C, Axes A-B, C-D, E-F | 3.2.5.1.2 | 4.6.10.1.2 | 1 - 7 |
| Random vibration | +57°C, Axes A-B, C-D, E-F | 3.2.5.1.3 | 4.6.10.1.3 | 1 - 7 |
| Random vibration | -43°C, Axes A-B, C-D, E-F | 3.2.5.1.3 | 4.6.10.1.3 | 1 - 7 |
| Temperature shock and altitude: | (Nonoperating preconditioning) 4/ | 3.2.5.1.4 | 4.6.10.1.4 | 1 - 7 |
| Handling shock: | (Nonoperating preconditioning) 4/ | 3.2.5.1.5 | 4.6.10.1.5 | |
| Test 1 | +57°C, Axes A-B, C-D, E-F | | | 1 - 7 |
| Test 2 | -43°C, Axes A-B, C-D, E-F | | | 1 - 7 |
| Humidity | (Nonoperating preconditioning) | 3.2.5.1.6 | 4.6.10.1.6 | 1 - 7 |
| Table II, Group 2 | Ambient conditions 2/ (4.3) | - - - - - | - - - - - | 1 - 7 |
| Terminal/header strength | | 3.2.1.10 | 4.6.7.4 | 1 - 7 |
| Leakage | Fine (4.4.8.1) | 3.2.2.3 | 4.6.8.1 | 1 - 7 |
| | Gross (4.4.8.2) | 3.2.2.3 | 4.6.8.2 | |
| Caged restriction | | 3.2.1.2.1 | 4.6.2.1 | 1 - 7 |



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TABLE I. First article inspection. - Continued.

| Inspection requirement 5/ | Conditions and reference paragraph (4.6) | Requirement paragraph | Inspection paragraph | Specimen Nos. 1/ 3/ |
|---|---|---|---|---|
| Actuation 1: | -43°C | 3.2.1.3.3 | 4.6.3 | 1 - 7 |
| 1. Uncaged indication | | 3.2.1.3.4 | 4.6.3.4 | |
| 2. Outer gimbal freedom | | 3.2.1.6.2 | 4.6.4.3 | |
| 3. Sine and cosine approximation | | 3.2.1.7.2 | 4.6.5.2 | |
| 4. Uncaging offset | | 3.2.1.7.3 | 4.6.5.3 | |
| 5. Inner gimbal lock indication | | 3.2.1.6.1.1 | 4.6.4.2 | |
| 6. Actuator resistance (post-fire) | | 3.2.1.3.1 | 4.6.3.1 | |
| 7. Actuator insulation resistance | | 3.2.1.9.3 | 4.6.7.3 | |
| Actuation 2: | -43°C, Fast roll, CCW (4.6.6.1) | | | 1, 3, 5, 7 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Fast roll drift | | 3.2.1.8 | 4.6.6.1 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 2: | -43°C, Slow roll, CCW (4.6.6.2) | | | 2, 4, 6 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Slow roll drift | | 3.2.1.8 | 4.6.6.2 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 3: | -43°C, Fast roll, CW | | | 1, 3, 5, 7 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Fast roll drift | | 3.2.1.8 | 4.6.6.1 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |



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TABLE I. First article inspection. - Continued.

| Inspection requirement 5/ | Conditions and reference paragraph (4.6) | Requirement paragraph | Inspection paragraph | Specimen Nos. 1/ 3/ |
|---|---|---|---|---|
| Actuation 3: | -43°C, Slow roll, CW | | | 2, 4, 6 |
| Repeat items 1, 2, and 5 under actutation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Slow roll drift | | 3.2.1.8 | 4.6.6.2 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 4: | -43°C, Slow roll, CCW | | | 1, 3, 5, 7 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Slow roll drift | | 3.2.1.8 | 4.6.6.2 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 4: | -43°C, Fast roll, CW | | | 2, 4, 6 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Fast roll drift | | 3.2.1.8 | 4.6.6.1 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 5: | -43°C | | | 1 thru 7 3/ |
| Repeat items 1 thru 4 under actuation 1. | | | | |
| Actuation 6: | -43°C, Slow roll, CW | | | 1, 3, 5, 7 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Slow roll drift | | 3.2.1.8 | 4.6.6.2 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |



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TABLE I. First article inspection. - Continued.

| Inspection requirement 5/ | Conditions and reference paragraph (4.6) | Requirement paragraph | Inspection paragraph | Specimen Nos. 1/ 3/ |
|---|---|---|---|---|
| Actuation 6: | -43°C, Fast roll, CCW | | | 2, 4, 6 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Fast roll drift | | 3.2.1.8 | 4.6.6.1 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 7: | +57°C, Fast roll, CCW | | | 1, 2, 3 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Fast roll drift | | 3.2.1.8 | 4.6.6.1 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 7: | +57°C, Slow roll, CCW | | | 5, 6, 7 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Fast roll drift | | 3.2.1.8 | 4.6.6.1 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 8: | +57°C, | | | 1 thru 7 3/ |
| Repeat items 1 thru 3 under actuation 1. | | | | |
| Actuation 9: | +57°C, Fast roll, CW | | | 1, 2, 6, 7 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Fast roll drift | | 3.2.1.8 | 4.6.6.1 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |



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TABLE I. First article inspection. - Continued.

| Inspection requirement 5/ | Conditions and reference paragraph (4.6) | Requirement paragraph | Inspection paragraph | Specimen Nos. 1/ 3/ |
|---|---|---|---|---|
| Actuation 9: | +57°C, Slow roll, CW | 3.2.1.3.3 | 4.6.3 | 3, 4, 5 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Slow roll drift | | 3.2.1.8 | 4.6.6.2 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 10: | +57°C, Slow roll, CCW | | | 1, 2, 3, 4 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Slow roll drift | | 3.2.1.8 | 4.6.6.2 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 10: | +57°C, Fast roll, CCW | | | 5, 6, 7 |
| Repeat items 1, 2, and 5 under actuation 1. | | | | |
| 1. Runup time | | 3.2.1.4 | 4.6.3.5 | |
| 2. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 | |
| 3. Fast roll drift | | 3.2.1.8 | 4.6.6.1 | |
| 4. Running time | | 3.2.1.5 | 4.6.3.6 | |
| Actuation 11: | Acceleration - ambient | | | 1 thru 7 3/ |
| Repeat items 1, 4, and 5 under actuation 1. | | 3.2.5.2.2 | 4.6.10.2.2 | |
| Actuation 12: | +57°C | | | 1 thru 7 3/ |
| Repeat items 1 thru 5 under actuation 1. | | | | |

NOTES:

1/ Specimens 1, 2, 3, 4, 5, 6, and 7 shall be subjected to those tests specified prior to primary sealing.



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NOTES: - Continued.

2/ Except perform the actuator no-fire current test at the gyro temperature of not less than +57°C.

3/ Specimen 8 set aside as reserve against accidental loss of any other specimen. To substitute for a lost specimen, No. 8 shall first undergo stresses accumulated by lost specimen to point of loss. If specimen 8 is not subjected to any of the stresses or tests of lot acceptance tests (LAT), it shall be returned to the lot from which it was selected.

4/ The inner gimbal lock indication switch shall be monitored as specified in 4.6.4.2 during the sine vibration and random vibration testing of the nonoperation preconditioning test specified herein. A momentary closure of the inner gimbal lock indication switch during any one of these tests shall constitute failure of the first article inspection.

5/ Procedures to re-establish the gyro functions between actuation shall be submitted to and approved by the contracting activity.

TABLE II. Gyro quality conformance inspection.

| Inspection or test | Requirement paragraph | Inspection paragraph |
|---|---|---|
| Group 1 1/ | | |
| Manual | | |
| Actuation 1 (Ambient, CW, Slow roll): | | |
| a. Uncaged indication | 3.2.1.3.4 | 4.6.3.4 |
| b. Running time | 3.2.1.5 | 4.6.3.6 |
| c. Inner gimbal lock indication | 3.2.1.6.1.1 | 4.6.4.2 |
| d. Slow roll drift | 3.2.1.8 | 4.6.6.2 |
| e. Uncaging offset | 3.2.1.7.3 | 4.6.5.3 |
| Manual | | |
| Actuation 2 (Ambient, CCW, Fast roll): | | |
| a. Uncaged indication | 3.2.1.3.4 | 4.6.3.4 |
| b. Running time | 3.2.1.5 | 4.6.3.6 |
| c. Inner gimbal lock indication | 3.2.1.6.1.1 | 4.6.4.2 |
| d. Fast roll drift | 3.2.1.8 | 4.6.6.1 |
| e. Uncaging offset | 3.2.1.7.3 | 4.6.5.3 |



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TABLE II. Gyro quality conformance inspection. - Continued

| Inspection or test | Requirement paragraph | Inspection paragraph |
|---|---|---|
| Group 2 2/ | | |
| Actuator no-fire current | 3.2.1.3.2 | 4.6.3.2 |
| Actuator resistance (pre-fired) | 3.2.1.3.1 | 4.6.3.1 |
| Caged indication (present) | 3.2.1.2.2 | 4.6.2.2 |
| Uncaged indication (absent) | 3.2.1.3.4 | 4.6.3.4 |
| Inner gimbal lock indication (absent) | 3.2.1.6.1.1 | 4.6.4.2.1 |
| Pickoff excitation resistance | 3.2.1.7.1 | 4.6.5.1 |
| Dielectric withstanding voltage | 3.2.1.9.1 | 4.6.7.1 |
| Insulation resistance | 3.2.1.9.2 | 4.6.7.2 |
| Group 3 3/ (Primary sealed gyros only.) | | |
| Weight and dimensions | 3.2.2.1, 3.2.2.2 | 4.6.1(d) |
| Identification and marking | 3.3.2 | 4.6.1(g) |
| Workmanship (exterior) | 3.5 | 4.6.1(h) |
| Cleanliness | 3.3.4.1 | 4.6.11(c) |
| Group 4 4/ (Primary sealed gyros only.) | | |
| Fine leak test | 3.2.2.3 | 4.6.8.1 |
| Gross leak test | 3.2.2.3 | 4.6.8.2 |
| Group 5 5/ | | |
| Reliability | 3.2.3 | 4.6.9 |
| Life | 3.2.3.1 | 4.6.9.1 |
| Group 6 4/ | | |
| Packaging | 5 | 4.6.1(i) |

NOTES:

1/ Tests to be conducted prior to the primary seal operation.

2/ Verification by gyro test at ambient conditions. For the actuator no-fire current test use gyro temperature of not less than +57 ±5°C.

3/ Verification by visual examination of gyro or examination of manufacturing data and manufacturer's certifications, except weight measures.

4/ Verification by visual examination of gyro packaging and packing.

5/ Reliability and life shall not be part of acceptance testing.



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4.5.2 Lot inspection. Lot sample gyros shall be randomly selected from each lot offered for acceptance in accordance with MIL-STD-105 and shall be tested in the sequence of Table III. The sequence for the performance of +57°C random vibration, -43°C random vibration, +57°C handling shock and -43°C handling shock tests is optional, although they shall occur between the Table II, Group 2, sets of measurements.

4.5.2.1 Inspection lot description. An inspection lot shall be as described in MIL-STD-105.

TABLE III. Lot inspection.

| Inspection or test 3/ | Condition and reference paragraph (4.6) | Requirement paragraph | Inspection paragraph |
|---|---|---|---|
| Table II, Groups 1, 2, 3 and 4 inspection | Ambient temperature conditions (see 4.3) | Table II | Table II |
| Temperature cycling and vibration temperature cycle: | Nonoperating, preconditioning 2/ | 3.2.5.1.1 | 4.6.10.1.1 |
| a. Random vibration | +57°C, Axes A-B, C-D, E-F | 3.2.5.1.3 | 4.6.10.1.3 |
| b. Random vibration | -43°C, Axes A-B, C-D, E-F | 3.2.5.1.3 | 4.6.10.1.3 |
| Handling shock: | Nonoperating, preconditioning | | |
| a. Test 1 | +57°C, Axes A-B, C-D, E-F | 3.2.5.1.5 | 4.6.10.1.5 |
| b. Test 2 | -43°C, Axes A-B, C-D, E-F | 3.2.5.1.5 | 4.6.10.1.5 |
| Table II, Group 2 | Ambient temperature conditions (see 4.3) | Table II | Table II |
| Terminal/header strength | | 3.2.1.10 | 4.6.1(c) |
| Leakage | Fine<br>Gross (4.4.8.2) | 3.2.2.3<br>3.2.2.3 | 4.6.8.1<br>4.6.8.2 |



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TABLE III. Lot inspection. - Continued.

| Inspection or test 3/ | Condition and reference paragraph (4.6) | Requirement paragraph | Inspection paragraph |
|---|---|---|---|
| Actuation 1: | -43°C, Fast roll, CW 1/ (see 4.4.6.1) | 3.2.1.3.3 | 4.6.3 |
| a. Uncaged indication | | 3.2.1.3.4 | 4.6.3.4 |
| b. Runup time | | 3.2.1.4 | 4.6.3.5 |
| c. Outer gimbal freedom | | 3.2.1.6.2 | 4.6.4.3 |
| d. Noise spikes | | 3.2.1.7.2 | 4.6.5.2 |
| e. Running time | | 3.2.1.5 | 4.6.3.6 |
| f. Inner gimbal lock indication | | 3.2.1.6.1.1 | 4.6.4.2 |
| g. Actuator resistance (post-fire) | | 3.2.1.3.1 | 4.6.3.1 |
| h. Actuator insulation resistance | | 3.2.1.9.3 | 4.6.7.3 |
| i. Fast roll drift | | 3.2.1.8 | 4.6.6.1 |
| Actuation 2: | Ambient (see 4.3) | | |
| a. Uncaged indication | | 3.2.1.3.4 | 4.6.3.4 |
| b. Outer gimbal freedom | | 3.2.1.6.2 | 4.6.4.3 |
| c. Sine and cosine approximation | | 3.2.1.7.2 | 4.6.5.2 |
| d. Uncaging offset | | 3.2.1.7.3 | 4.6.5.3 |
| e. Inner gimbal lock indication | | 3.2.1.6.1.1 | 4.6.4.2 |
| Actuation 3: | +57°C, Slow roll, CCW (see 4.6.6.2) | | |
| Repeat items 1 thru 6 under actuation 1. | | | |
| 9. Slow roll drift | | 3.2.1.8 | 4.6.6.2 |
| Actuation 4: | Ambient, Fast roll, CCW (see 4.6.6.1) | | |
| Repeat items 1 thru 6 and 9 under actuation 1. | | | |



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TABLE III. Lot inspection. - Continued.

| Inspection or test 3/ | Condition and reference paragraph (4.4) | Requirement paragraph | Inspection paragraph |
|---|---|---|---|
| Actuation 5: | Ambient, Slow roll, CW (see 4.4.6.2) | | |
| Repeat items 1 thru 6 under actuation 1. | | | |
| 9. Slow roll drift | | 3.2.1.8 | 4.6.6.2 |
| Actuation 6: | Ambient (see 4.3) | | |
| Repeat items 1 thru 5 under actuation 2. | | | |

NOTES:

1/ Specimens 1, 2, 3, 4, 5, 6, and 7 shall be subjected to those tests specified after being primary sealed. Specimen 8 set aside as reserve against accidental loss of any other specimen. To substitute for a lost specimen, No. 8 shall first undergo stresses accumulated by lost specimen to point of loss.

2/ The inner gimbal lock indication switch shall be monitored as specified in 4.6.4.2 during the random vibration testing of the nonoperating preconditioning tests specified. A momentary closure of the inner gimbal lock indication switch during any one of these tests shall constitute failure of the lot inspection.

3/ See 5/ of Table I.

4.5.2.2 Lot sampling. After being subjected to Group 1 of the individual tests of Table II, a lot sample of eight gyros shall be randomly selected from the lot. The quantity in a lot shall be not greater than 150. The samples shall be selected prior to primary sealing of the entire lot. The lot samples shall be primary sealed and tested in accordance with Table III. The remaining units of the lot shall be sealed and tested in accordance with Table II, Groups 2, 3 and 4, after successful completion of lot acceptance tests (LAT).

4.5.2.3 Action in the event of LAT failure. Failure of the sample to meet the acceptance criteria of 4.5.2.2 shall be cause for rejection of the lot. The contractor shall then take action acceptable to the contracting activity to correct the cause(s) of failure.



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4.5.2.4 Disposition of test samples. LAT samples shall be refurbished and submitted for acceptance with a subsequent lot. The final lot sample shall be refurbished and delivered upon approval from the contracting activity.

4.6 Test and inspection methods. The following methods shall be applied in inspections at the times and under the circumstances indicated in Tables I, II and III. The tests specified in Tables I, II or III which are performed prior to the primary seal shall be conducted in an environment which meets the sealing requirements obtained by the secondary seal. Sealing shall be accomplished by using the secondary cover seal, with suitable means to attach it to the gyro base, or by utilization of test fixturing. In either case, the gyro shall be conditioned and tested in an environment of dry nitrogen or a mixture of dry nitrogen percent and helium percent. During the actuation sequences of Tables I and III, the first actuation of each gyro shall utilize the actuation mechanism specified in 3.2.1.3. All subsequent actuations shall be initiated using an electro-mechanical or mechanical mechanism that is an integral part of the test fixture and that maintains the seal requirements specified for the test sequence. Removal of the gyro from its secondary sealed environment will be permitted between actuations for the purposes of resetting the actuating mechanism and rewinding the motor spring. These operations shall take place in a clean environment as specified in 4.3.2. The tests specified in Tables I, II and III which require verification of the uncaged indication, inner gimbal lock indication, runup time and running time during an actuation sequence shall be monitored using a test verification circuit meeting the requirements of the schematic shown on Figure 3.

4.6.1 Visual and mechanical inspection. Visually and with mechanical gagings as necessary, the gyro and its preparation for delivery shall be examined for compliance with the following:

| | Item | Requirement paragraph |
|---|---|---|
| a. | Motor winding: | 3.2.1.1 |
| b. | Caging restrictions: | 3.2.1.2.1 |
| c. | Terminal or header pin strength: | 3.2.1.10 |
| d. | Weight and dimensions: | 3.2.2.1, 3.2.2.2 |
| e. | Life: | 3.2.3.1 |
| f. | Materials: | 3.3.1.1 |
| g. | Identification and marking: | 3.3.2 |



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| Item | Requirement paragraph |
|---|---|
| h. Workmanship: | 3.5 |
| i. Preservation, packaging and packing: | 5.1 and 5.2 |
| j. Marking for shipment and storage: | 5.3 |

4.6.2 Caging tests.

4.6.2.1 Caged restriction. With the gyro caged and wound as in 3.2.1.1, the total angular play about the outer gimbal axis shall be not greater than 1 degree when the outer gimbal is rotated alternately CW and CCW in accordance with 3.2.1.2.1.

4.6.2.2 Caged indication. With the gyro caged and the motor fully wound, a resistance of not greater than 1.0 ohm, measured at not greater than 10 mAdc, shall appear between terminals 3 and 4. A resistance of not less than 10 megohms, measured at 100 Vdc, shall appear between terminals 2 and 4 in accordance with 3.2.1.2.2.

4.6.3 Actuation. The electrical characteristics and operation of the gyro actuator mechanism shall meet the requirements of 3.2.1.3 when tested as specified in 4.6.3.1 through 4.6.3.6.

4.6.3.1 Pre-fired/post-fired actuator resistance. The resistance between terminals 4 and 5 shall be measured using not greater than 10 mAdc current and with the gyro at 25 ±5°C. The resistance shall meet the requirements of 3.2.1.3.1.

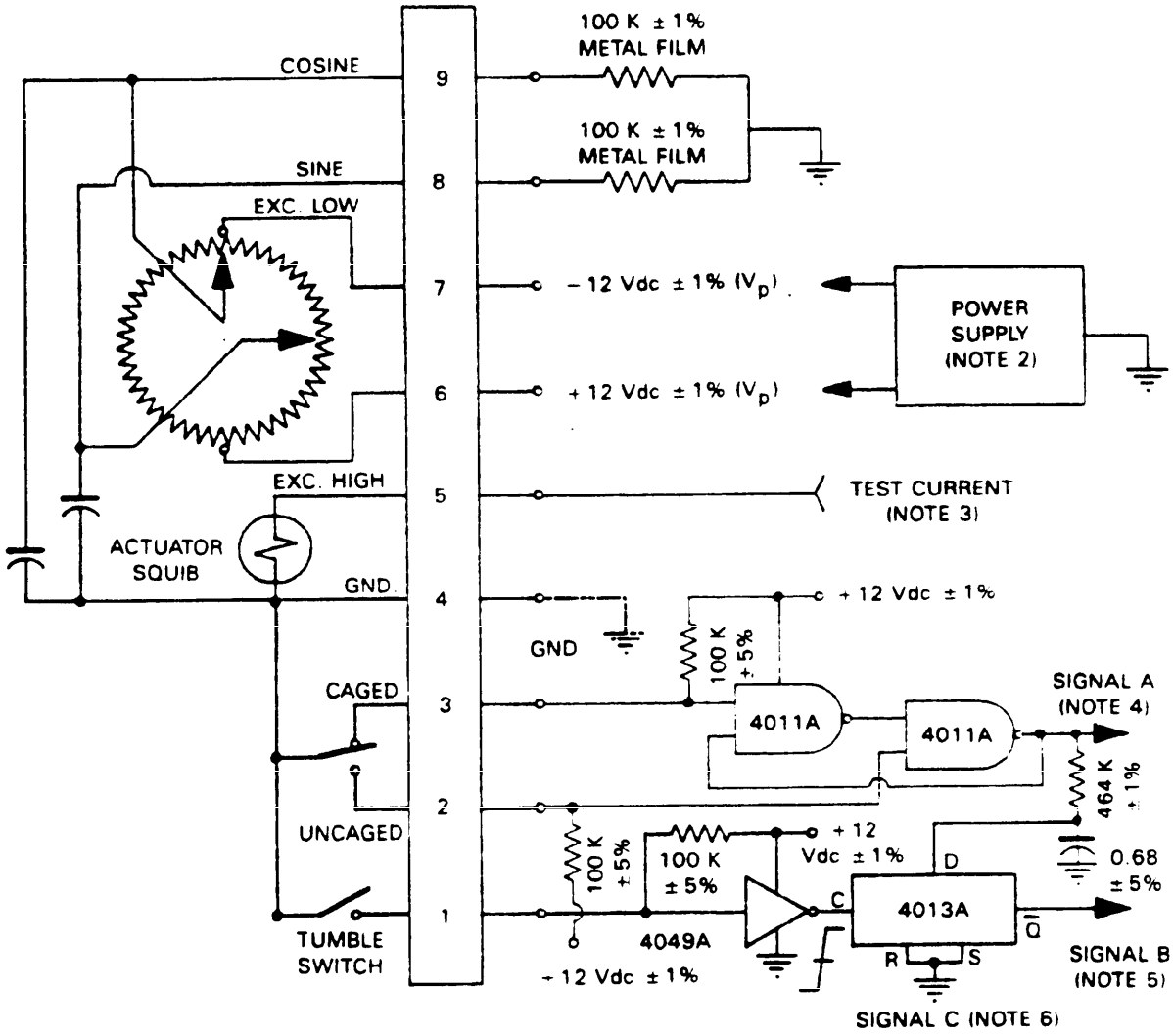
4.6.3.2 Actuator no-fire current. With the gyro stabilized at 57 ±5°C, a current of 30 +0, -3 mAdc shall be applied between terminals 4 and 5 for a period of not less than 5 minutes. Under this condition the actuator shall meet the requirements of 3.2.1.3.2.

4.6.3.3 Actuator firing current. A current of 300 +10, -0 mAdc shall be applied between terminals 4 and 5 for a period of not greater than 10 ms. Under this condition, the actuator shall release the spring motor in accordance with 3.2.1.3.3.

4.6.3.4 Uncaged indication. The resistance shall be monitored between terminals 2 and 4, and terminals 3 and 4 from the onset of the current of 3.2.1.3.3 until the gimbal lock indicaton of 3.2.1.6.1.1. The gyro shall meet the requirements of 3.2.1.3.4 during this interval. The gyro test circuit of Figure 3 may be used to verify timing of both the caged/uncaged indication and the gimbal lock indication. Resistance values as specified herein shall be verified prior to and subsequent to an individual test.



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NOTES:
1. SINE AND COSINE OUTPUTS ARE BOTH REFERRED TO GROUND, PIN 4.
2. TOLERANCE OF THE ± 12 Vdc POWER SUPPLY SHALL BE ± 1 PERCENT.
3. THE TEST CURRENT SHALL MEET THE REQUIREMENTS OF 3.2.1.3.3.
4. SIGNAL A IS 0 FOR CAGED AND 1 FOR UNCAGED.
5. SIGNAL B IS 1 FOR G-BIAS ENABLE.
6. SIGNAL C IS 0 FOR NO TUMBLE AND 1 FOR TUMBLE.

FIGURE 3. Gyro test circuit.



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4.6.3.5 Runup time. With the gyro actuated in accordance with 4.6.3.3, verification shall be made that the gyro meets the requirements of 3.2.1.4.

4.6.3.6 Running time. Gyro running time shall be measured with the gyro outer gimbal axis in the horizontal plane within ±10 degrees. The roll rate specified in 3.2.1.5 shall be established within not greater than 2.0 seconds after receipt of the uncaged indication of 3.2.1.3.4. The actuator firing current shall be applied with the gyro rotating at 0.0 ±10 deg per sec and the inner gimbal axis vertical. The gyro running time to gimbal lock shall meet the requirements of 3.2.1.5.

4.6.4 Gimbal freedom.

4.6.4.1 Inner gimbal freedom measurement. Inner gimbal freedom shall be measured with the gyro cover removed. The inner gimbal shall be rotated both CW and CCW about its rotation axis. All measurements shall comply with 3.2.1.6.1.

4.6.4.2 Inner gimbal lock indication. The resistance between terminals 1 and 4 shall be monitored to verify that the inner gimbal lock indication meets the requirements of 3.2.1.6.1.1. The gyro testing circuit on Figure 3 shall be used to verify both the timing and the resistance of the momentary closure of the inner gimbal lock indicator.

4.6.4.2.1 Inner gimbal lock indication (absence). Verification shall be made that the inner gimbal lock indication is absent by measuring the resistance between terminals 1 and 4 at 100 Vdc. This resistance shall meet the requirements of 3.2.1.6.1.1 for the inner gimbal within limits of angular motion of 3.2.1.6.1.

4.6.4.3 Outer gimbal freedom. The gyro shall be tested to ensure that the outer gimbal freedom of motion meets the requirements of 3.2.1.6.2.

4.6.5 Pickoff potentiometer.

4.6.5.1 Excitation resistance. Under the conditions specified in 3.2.1.7.1, the resistance shall be measured between terminals 6 and 7 with terminals 8 and 9 unconnected and the gyro at 25 ±5°C. The measured resistance shall meet the requirements of 3.2.1.7.1.

4.6.5.2 Sine and cosine approximation measurement. With the pickoff potentiometer excited, as shown in Figure 1, the gyro shall meet the requirements of 3.2.1.7.2. The gyro test circuit shown on Figure 3 shall excite the potentiometer as required. Gyro case rotation rate shall be 10 deg per sec 20 deg per sec unless otherwise specified in the conditions and reference columns of the applicable inspection table.

4.6.5.3 Uncaging offset. The gyro shall be tested to ensure that the uncaging offset requirements of 3.2.1.7.3 are met.



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4.6.6 Drift. Gyro drift with roll shall be measured with the gyro outer gimbal axis in the horizontal plane within ±10 degrees. The roll rate as specified in either 4.6.6.1 or 4.6.6.2, as appropriate for the test being performed, shall be established within not greater than 2.0 seconds after receipt of the uncaged indication of 3.2.1.3.4. The actuator firing current shall be applied with the gyro rotating at 0.0 ±10 deg per sec with the inner gimbal axis vertical. Gyro drift with roll shall meet the requirements of 3.2.1.8 when tested in accordance with 4.6.6.1 and 4.6.6.2.

4.6.6.1 Fast roll. The gyro case shall be accelerated to 260 ±10 deg per sec rotation in either direction about the outer gimbal axis alone. The drift measured after the running time ±5 seconds specified in 3.2.1.5 shall be not greater than 20 degrees.

4.6.6.2 Slow roll. The gyro case shall be accelerated to 75 ±5 deg per sec rotation in either direction about the outer gimbal axis alone. The drift measured after the running time ±5 seconds specified in 3.2.1.5 shall be not greater than 20 degrees.

4.6.7 Electrical tests.

4.6.7.1 Dielectric withstanding voltage. Upon application of 200 Vrms, 60 Hz, between the gyro terminal under test and gyro case in accordance with MIL-STD-202, Method 301, and with terminals 4 and 5 jumpered, the gyro shall meet the requirements of 3.2.1.9.1.

4.6.7.2 Insulation resistance. With terminals 4 and 5 jumpered, the insulation resistance shall be measured between electrically isolated interior circuits and the gyro case in accordance with MIL-STD-202, Method 302, Test Condition A. The gyro shall meet the requirements of 3.2.1.9.2.

4.6.7.3 Actuator insulation resistance. The resistance between terminal 5 and the gyro case shall be measured at not greater than 10 mAdc. The gyro shall meet the requirements of 3.2.1.9.3.

4.6.7.4 Terminal/header strength. The terminals (Drawing 1563560) or header pins (Drawing 1563567) shall be verified to meet the requirements of 3.2.1.10 and the applicable drawing.

4.6.8 Leakage test. To ensure the gyro meets the requirements of 3.2.2.3, leakage measurements shall be made in accordance with MIL-STD-202, Method 112, Condition B and C. The fine leak test of 4.6.8.1 shall not be considered successful unless followed by the successful completion of 4.6.8.2.

4.6.8.1 Fine leak test. A fine leak test shall be performed in accordance with MIL-STD-202, Method 112, Procedure IV, Condition C.



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4.6.8.2 Gross leak test. Gross leak test shall be performed in accordance with MIL-STD-202, Method 112, Condition B, and water may be substituted for silicone oil. A vacuum may be simulated by increasing the fluid temperature so the volume of gas sealed in the room-temperature gyro increases in pressure as the trapped gases expand due to the heating of the warmed fluid (silicone oil or water). The appropriate temperature versus gyro gas volume shall be calculated or measured and justified prior to the use of the hot water alternate method. This bubble test shall be performed to verify the fine leak test.

4.6.9 Reliability. Reliability shall be demonstrated by completing the first article inspection of 4.4 with no failures. Subsequent to first article inspection, reliability shall be maintained by control of materials, parts, and processes and verified by the quality conformance inspection of 4.5 and in accordance with 3.2.3.

4.6.9.1 Life. The gyro shall be tested to ensure it meets the life requirements of 3.2.3.1.

4.6.10 Environmental test.

4.6.10.1 Nonoperating environments. The gyro shall meet the nonoperating environmental requirements of 3.2.5.1 when tested as specified in 4.6.10.1.1 through 4.6.10.1.6.

4.6.10.1.1 Temperature cycle. Gyro temperature shall be raised within 1 hour to +57 ±5°C, held at that high level for not less than 4 hours, lowered within 2 hours to -43 ±5°C, held at that low level for not less than 4 hours, and returned within 1 hour to room ambient. This constitutes the first of two such temperature cycles that shall be carried out. When required by Table I or III, the sine vibrations of 3.2.5.1.2 shall be applied during the high and again during the low level dwell of the first temperature cycle, and the random vibrations of 3.2.5.1.3 shall be similarly applied during the second temperature cycle. The temperature cycle shall be verified in accordance with 3.2.5.1.1.

4.6.10.1.2 Sine vibration. Sine vibration frequency shall be swept at a logarithmic rate from 5 to 500 to 5 Hz in 15 minutes. Vibration level at the gyro attachment points, within ±10 percent of acceleration or amplitude, shall be ±1.3 g peak, 5 to 26 Hz; 0.036 inch double amplitude, 26 to 52 Hz; and ±5 g peak, 52 to 500 Hz. Time per application shall be not less than four frequency sweeps (1 hour total) per axis in each of 3 axes (see Figure 2). The gyro shall meet the sine vibration requirements of 3.2.5.1.2.



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4.6.10.1.3 Random vibration. Random vibration power spectral density at the gyro attachment points shall be 0.010 $g^2$ per Hz ±3 dB from 20 to 2,000 Hz, with not less than 6 to 12 dB per octave rolloff below 20 and above 2,000 Hz. Time per application shall be not less than 1 hour per axis in each of 3 axes (see Figure 2). The gyro shall meet the random vibration requirements of 3.2.5.1.3.

4.6.10.1.4 Temperature shock and altitude. The gyro shall be held not less than 1 hour in a hot chamber at +71 ±5°C and ambient barometric pressure, transferred within not greater than 5 minutes to and held for not less than 1 hour in a cold chamber at -43 ±5°C and not less than 50,000-foot barometric altitude, and returned within not greater than 5 minutes to the hot chamber. This constitutes the first of six such cycles that shall be carried out. At the end of the last cycle the gyro shall remain in the hot chamber not less than 1 hour before being cooled to room temperature. The gyro shall meet the temperature shock and altitude requirements of 3.2.5.1.4.

4.6.10.1.5 Handling shock. Handling shock shall be half-sine wave shocks of not less than 20 ms duration at not less than 25 g peak amplitude, applied three times per direction in each direction along each of three axes (see Figure 2) with the gyro at +57 ±5°C, and reapplied similarly with the gyro at -43 ±5°C, for an overall total of 36 shocks. The gyro shall meet the handling shock requirements of 3.2.5.1.5.

4.6.10.1.6 Humidity. The gyro shall be tested for moisture resistance in accordance with MIL-STD-202, Method 106, with vibration omitted. The gyro shall meet the humidity requirements of 3.2.5.1.6.

4.6.10.2 Operating environments.

4.6.10.2.1 Temperatures. The gyro shall be actuated in accordance with 4.6.3.3 with the gyro at a temperature of -43 ±5°C and +57 ±5°C, respectively. The gyro shall meet the requirements of 3.2.5.2.1 under each condition.

4.6.10.2.2 Acceleration. The gyro shall operate as specified when exposed to 25 ±3 g acceleration applied along the A-B axis only (see Figure 2), in the direction A to B, for not less than 5 seconds starting at the onset of the voltage of 4.6.3.3. The gyro shall meet the requirements of 3.2.5.2.2.



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4.6.11 Data/certification inspection. Objective manufacturing data and contractor certifications shall be examined during collective inspection of a first article or periodic sample or first article lot of gyros for compliance with the following:

| Item | Requirement paragraph |
|---|---|
| a. Filling: | 3.2.2.3 |
| b. Materials, processes and parts: | 3.3.1 |
| c. Cleanliness: | 3.3.4.1 |

## 5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified in the contract or purchase order (see 6.2.1), preservation and packaging shall be in accordance with 5.1.1.

5.1.1 Level C. Preservation, packaging, and packing shall be in accordance with the contractor's commercial practices to afford protection from damage during shipment, within the scope of this specification, from the supply source to the first receiving activity.

5.2 Packing. Unless otherwise specified in the contract or purchase order (see 6.2.1) packing shall be in accordance with 5.2.1.

5.2.1 Level C. Gyros packaged as specified herein shall be packed in containers of the type, size, and kind commonly used for this purpose, in a manner that will ensure acceptance by common carrier and safe delivery at destination. Shipping containers shall be in compliance with the Uniform Freight Classification Ratings, Rules, and Regulations or the ratings, rules, and regulations for other carriers, as applicable to the mode of transportation.

5.3 Marking for shipment and storage. Unless the contract or purchase order specifies otherwise (see 6.2.1), unit packages, intermediate packages, and exterior shipping container shall be marked in accordance with MIL-STD-129.

## 6. NOTES

6.1 Intended use. The gyro covered by this document is intended for use in the guidance system of AGM-45A/B-9A air-launched missile to implement the gravity bias function.



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against accidental loss of any other specimen. To substitute for a

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition requirements should specify the following:

a. Title, number, and date of this specification.

b. Identification and marking (see 3.3.2).

c. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a first article sample consisting of 16 gyros as specified in 4.4. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests and approval of the first article.

d. Responsibility for inspection and inspection facilities, if other than as in 4.1.

e. Level of preservation and packaging, if different than 5.1.

f. Level of packing, if other than as in 5.2.

g. Marking for shipment and storage, if different than 5.3.

h. Size of first article sample, if different than 4.4.

i. First article inspection, if different than 4.1.

j. First article test samples procured under this specification are to be considered destructively tested and therefore should not be included in the net deliverable hardware.

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

| Paragraph no. | Data requirement title | Applicable DID no. | Option |
|---|---|---|---|
| 4.1.2 | Reports, Test | DI-T-2072 | |
| 4.1.2 | Test Procedures | UDI-T-21347 | |

(Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)



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6.2.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 16 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Subject term (keyword) listing.

Actuator, (squib)
AGM 45A/B-9A Shrike G-bias missile.
Caged
Gyro
Indicator, caged and wound
Roll position indicating
Tumble switch
Uncaged

Preparing activity:
Navy - AS

(Project No: 1420-N300)



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3.2.5.2 Operating environments

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |
| | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
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